

BUSINESS ANALYTICS REPORT

PROJECT BASED LEARNING REPORT (PBL)

Customer Segmentation using RFM Analysis and K-Means Clustering in Python

MASTER OF BUSINESS ADMINISTRATION (MBA)

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DECLARATION

I hereby declare that the Project Based Learning (PBL) report titled “**Customer Buying Behaviour Analysis using RFM (Python)**” submitted by me is a bonafide record of the work carried out under the guidance of my faculty.

I further declare that this project report has been prepared by me and has not been submitted to any other institution or university for the award of any degree, diploma, or certificate. The information and data included in this report have been collected from authentic sources and have been used solely for academic purposes.

I take full responsibility for the originality and authenticity of the work presented in this report. Wherever necessary, proper acknowledgment has been given to the sources of information used.

This project has helped me gain practical knowledge of data analysis, customer segmentation, and the application of RFM analysis using Python. It has enhanced my understanding of real-world data science concepts and their business applications.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who helped me in the successful completion of this Project-Based Learning report.

First and foremost, I am deeply thankful to **Dr. Gaurav Chopra**, under whose guidance and support this project was successfully completed. His valuable insights, continuous encouragement, and expert guidance played a crucial role in shaping the direction of this project and enhancing my understanding of data analysis.

I would also like to thank my family and friends for their constant support and motivation throughout the duration of this project. Their encouragement helped me stay focused and complete the work efficiently.

I would like to extend my appreciation to all the online resources, datasets, and study materials that provided valuable information and practical knowledge. These resources greatly contributed to the successful execution of this project on **Customer Buying Behaviour Analysis using RFM (Python)**.

Lastly, I am grateful for the opportunity to work on this project, which has helped me gain practical exposure to data science concepts such as data cleaning, visualization, clustering, and classification.

EXECUTIVE SUMMARY

This project focuses on analyzing customer purchasing behavior using transaction data. The dataset was cleaned and processed using Python to ensure accuracy. RFM (Recency, Frequency, Monetary) analysis was performed to evaluate customer value and understand purchasing patterns.

Various visualizations such as bar charts, histograms, and line charts were used to identify trends in customer spending and sales over time. The results show that most customers contribute low revenue, while a small group of customers contributes significantly higher revenue

. K-Means clustering was applied to segment customers into different groups such as high-value, regular, and low-value customers. A classification model was also used to predict customer segments.

The analysis helps businesses understand customer behavior and design better marketing strategies to improve customer retention and profitability.

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OBJECTIVES OF THE STUDY

The main objectives of this project are:

- To analyze customer purchasing behavior using transaction data
- To perform RFM (Recency, Frequency, Monetary) analysis to evaluate customer value
- To visualize customer data using graphs and charts
- To segment customers into different groups using clustering techniques
- To apply classification methods to predict customer segments
- To derive meaningful insights that can help in improving business strategies

DATASET DESCRIPTION

The dataset used in this project is the **Online Retail dataset**, which contains transaction-level data of customers. It includes information about customer purchases such as invoice details, product quantity, price, and transaction dates.

The key variables in the dataset are:

- **CustomerID**: Unique identifier for each customer
- **InvoiceNo**: Unique transaction number
- **InvoiceDate**: Date and time of purchase
- **Quantity**: Number of items purchased
- **UnitPrice**: Price per unit of the product
- **TotalPrice**: Calculated as $\text{Quantity} \times \text{UnitPrice}$

The dataset contains real-world retail transaction data, making it suitable for analyzing customer purchasing behavior. It helps in performing RFM analysis and identifying valuable customers based on their buying patterns.

Before analysis, the dataset was cleaned by removing missing values and invalid records to ensure accuracy and reliability of results.

METHODOLOGY

The project follows a systematic approach to analyze customer purchasing behavior using data science techniques. The steps involved are as follows:

1. Data Collection

The dataset used is the Online Retail dataset, which contains transaction-level data of customers including purchase date, quantity, and price.

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom

- `df = pd.read_excel("E:/r code/pbl 3 data set rfm analysis R/Online Retail.xlsx")`
- `df.head()`
- `df.info()`
- `df.describe()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   InvoiceNo        541909 non-null object
1   StockCode       541909 non-null object
2   Description     540455 non-null object
3   Quantity        541909 non-null int64
4   InvoiceDate     541909 non-null datetime64[ns]
5   UnitPrice      541909 non-null float64
6   CustomerID     406829 non-null float64
7   Country        541909 non-null object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 33.1+ MB
```

	Quantity	InvoiceDate	UnitPrice	CustomerID
count	541909.000000	541909	541909.000000	406829.000000
mean	9.552250	2011-07-04 13:34:57.156386048	4.611114	15287.690570
min	-80995.000000	2010-12-01 08:26:00	-11062.060000	12346.000000
25%	1.000000	2011-03-28 11:34:00	1.250000	13953.000000
50%	3.000000	2011-07-19 17:17:00	2.080000	15152.000000
75%	10.000000	2011-10-19 11:27:00	4.130000	16791.000000
max	80995.000000	2011-12-09 12:50:00	38970.000000	18287.000000
std	218.081158	NaN	96.759853	1713.600303

2. Data Cleaning

The dataset was cleaned to ensure accuracy by:

- Removing missing values in the CustomerID column
- Removing negative quantities (returns or cancelled transactions)
- Creating a new variable, TotalPrice, calculated as $\text{Quantity} \times \text{UnitPrice}$

```
➤ df = df.dropna(subset=['CustomerID'])
➤ df = df[df['Quantity'] > 0]
➤ df['TotalPrice'] = df['Quantity'] * df['UnitPrice']
➤ df.head()
```

3. Data Sampling

A random sample of 5000 records was selected from the dataset to reduce computational complexity and make the analysis efficient.

```
➤ df = df.sample(5000, random_state=123)
➤ df.head()
```

4. RFM Analysis

RFM (Recency, Frequency, Monetary) analysis was performed to evaluate customer behavior:

- **Recency:** Number of days since last purchase
 - **Frequency:** Number of transactions made by the customer
 - **Monetary:** Total amount spent by the customer
- ```
➤ df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
➤ snapshot_date = df['InvoiceDate'].max() +
 pd.Timedelta(days=1)
➤ rfm = df.groupby('CustomerID').agg({
➤ 'InvoiceDate': lambda x: (snapshot_date - x.max()).days,
➤ 'InvoiceNo': 'count',
➤ 'TotalPrice': 'sum'
➤ })
➤ rfm.columns = ['Recency', 'Frequency', 'Monetary']
➤ rfm.head()
```

|            | Recency | Frequency | Monetary |
|------------|---------|-----------|----------|
| CustomerID |         |           |          |
| 12347.0    | 183     | 3         | 66.30    |
| 12349.0    | 19      | 1         | 12.50    |
| 12350.0    | 310     | 1         | 19.80    |
| 12356.0    | 326     | 1         | 75.00    |
| 12357.0    | 33      | 3         | 64.56    |

## RFM SCORING AND CUSTOMER SEGMENTATION

In addition to RFM analysis, an RFM scoring technique was applied to rank customers based on their purchasing behavior. Each customer was assigned a score from 1 to 5 for Recency, Frequency, and Monetary values.

- Recency score indicates how recently a customer made a purchase (lower recency = higher score)
- Frequency score indicates how often a customer makes purchases
- Monetary score indicates how much a customer spends

These scores were combined to form an overall RFM score for each customer. Based on this score, customers were segmented into different categories such as high-value, regular, and low-value customers.

This segmentation helps businesses identify their most valuable customers, target loyal customers, and take actions to re-engage inactive customers. The results show that a small group of customers fall under the high-value category, while the majority of customers are in the regular or low-value segment.

```

: # RFM Scoring (1-5 scale)

rfm['R_score'] = pd.qcut(rfm['Recency'], 5, labels=[5,4,3,2,1])
rfm['F_score'] = pd.qcut(rfm['Frequency'].rank(method='first'), 5, labels=[1,2,3,4,5])
rfm['M_score'] = pd.qcut(rfm['Monetary'], 5, labels=[1,2,3,4,5])

Combine scores
rfm['RFM_Score'] = rfm['R_score'].astype(str) + rfm['F_score'].astype(str) + rfm['M_score'].astype(str)

rfm.head()

```

```

: Recency Frequency Monetary R_score F_score M_score RFM_Score
CustomerID
12347.0 183 3 66.30 2 4 5 245
12349.0 19 1 12.50 5 1 2 512
12350.0 310 1 19.80 1 1 3 113
12356.0 326 1 75.00 1 1 5 115
12357.0 33 3 64.56 4 4 5 445

```

```

Segmentation

rfm['Segment'] = 'Regular'

rfm.loc[rfm['RFM_Score'] >= '444', 'Segment'] = 'High Value'
rfm.loc[rfm['RFM_Score'] <= '222', 'Segment'] = 'Low Value'

rfm[['Recency', 'Frequency', 'Monetary', 'RFM_Score', 'Segment']].head()

```

```

 Recency Frequency Monetary RFM_Score Segment
CustomerID
12347.0 183 3 66.30 245 Regular
12349.0 19 1 12.50 512 High Value
12350.0 310 1 19.80 113 Low Value
12356.0 326 1 75.00 115 Low Value
12357.0 33 3 64.56 445 High Value

```

## INTERPRETATION OF RFM SCORING

The RFM scoring analysis provides a detailed understanding of customer behavior by assigning scores based on Recency, Frequency, and Monetary values. Customers with higher RFM scores represent more valuable customers, as they purchase frequently, spend more, and have made recent transactions.

The analysis shows that a small group of customers have high RFM scores, indicating that they are loyal and high-value customers who contribute significantly to revenue. These customers are important for the business and should be retained through targeted marketing strategies and personalized offers.

Customers with medium RFM scores fall into the regular category. These customers have moderate purchasing activity and can be encouraged to increase their engagement through promotions and loyalty programs.

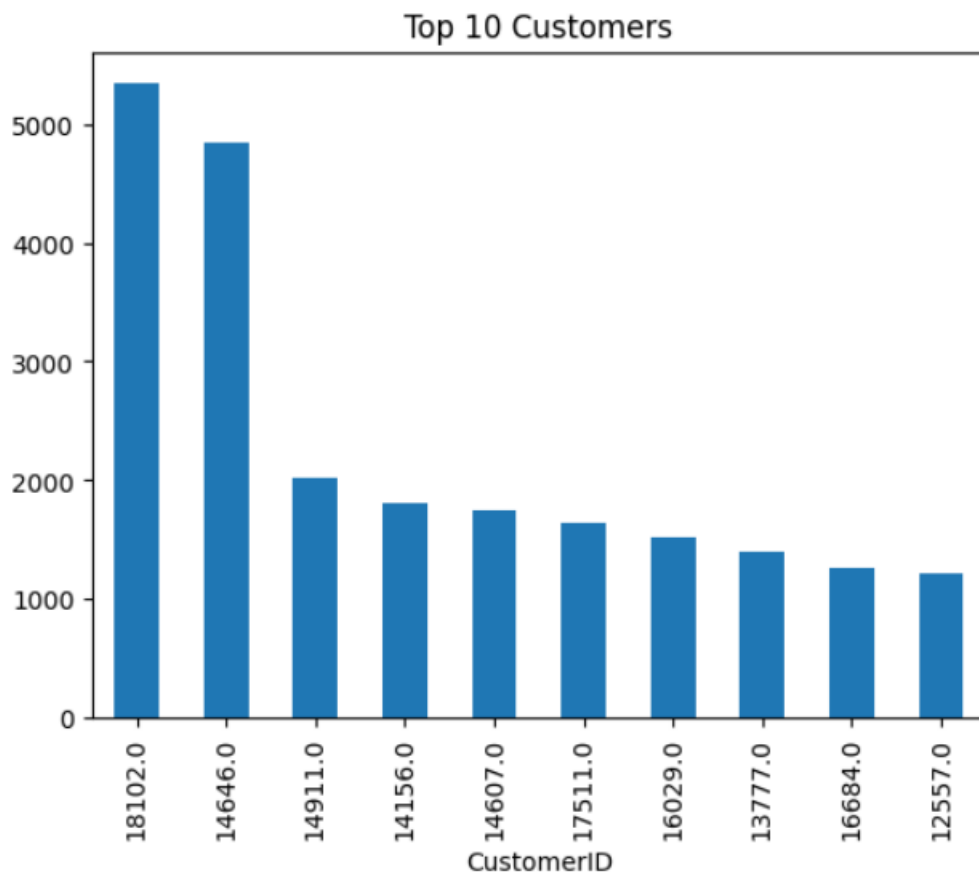
Customers with low RFM scores are less active and contribute less to overall revenue. These customers may require re-engagement strategies such as discounts or special campaigns to bring them back.

Overall, RFM scoring helps in ranking customers effectively and provides a clear basis for customer segmentation and decision-making.

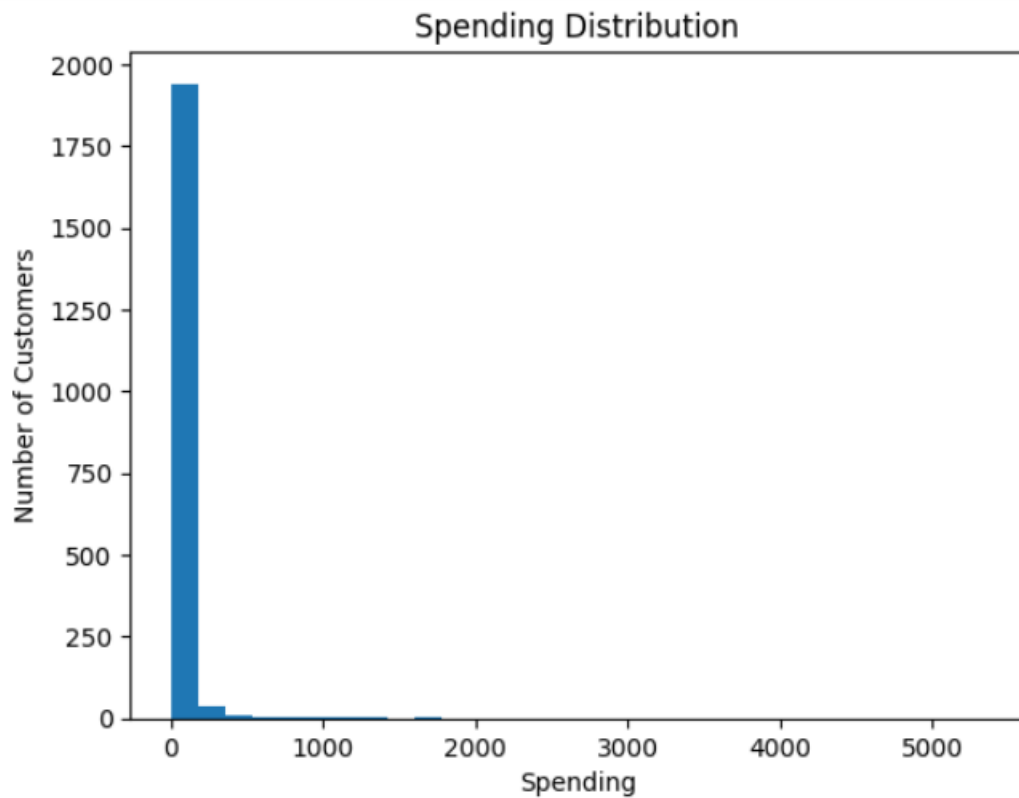
# Data Visualization

Various charts were used to understand patterns in the data:

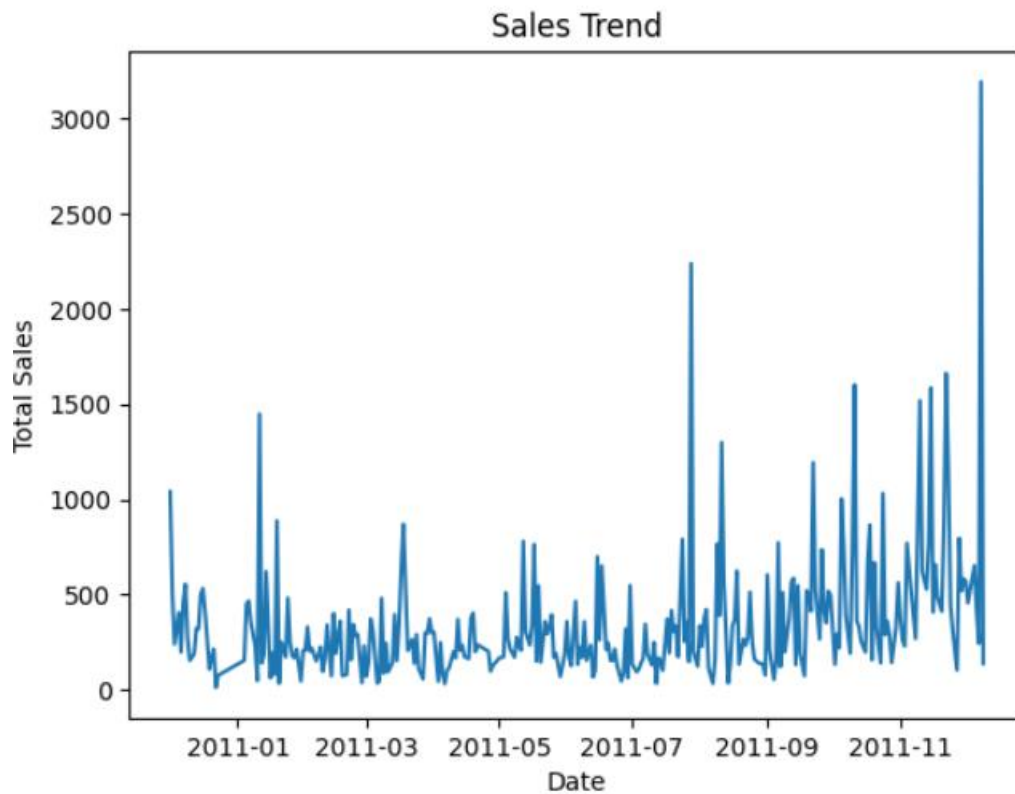
- Bar chart for top customers
  - `rfm.sort_values('Monetary', ascending=False).head(10)['Monetary'].plot(kind='bar')`
  - `plt.title("Top 10 Customers")`
  - `plt.show()`



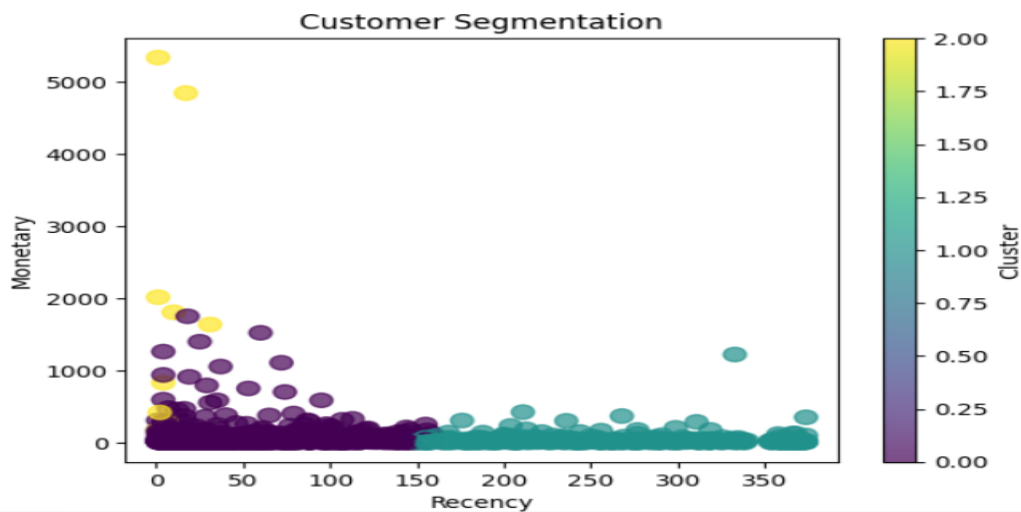
- Histogram for spending distribution
  - `plt.hist(rfm['Monetary'], bins=30)`
  - `plt.title("Spending Distribution")`
  - `plt.xlabel("Spending")`
  - `plt.ylabel("Number of Customers")`
  - `plt.show()`



- Line chart for sales trends
  - `sales = df.groupby(df['InvoiceDate'].dt.date)['TotalPrice'].sum()`
  - `sales.plot()`
  - `plt.title("Sales Trend")`
  - `plt.xlabel("Date")`
  - `plt.ylabel("Total Sales")`
  - `plt.show()`



- Scatter plot for customer segmentation
  - `plt.scatter(rfm['Recency'], rfm['Monetary'], c=rfm['Cluster'], cmap='viridis', s=80, alpha=0.7)`
  - `plt.title("Customer Segmentation")`
  - `plt.xlabel("Recency")`
  - `plt.ylabel("Monetary")`
  - # Adding color bar
  - `plt.colorbar(label='Cluster')`
  - `plt.show()`



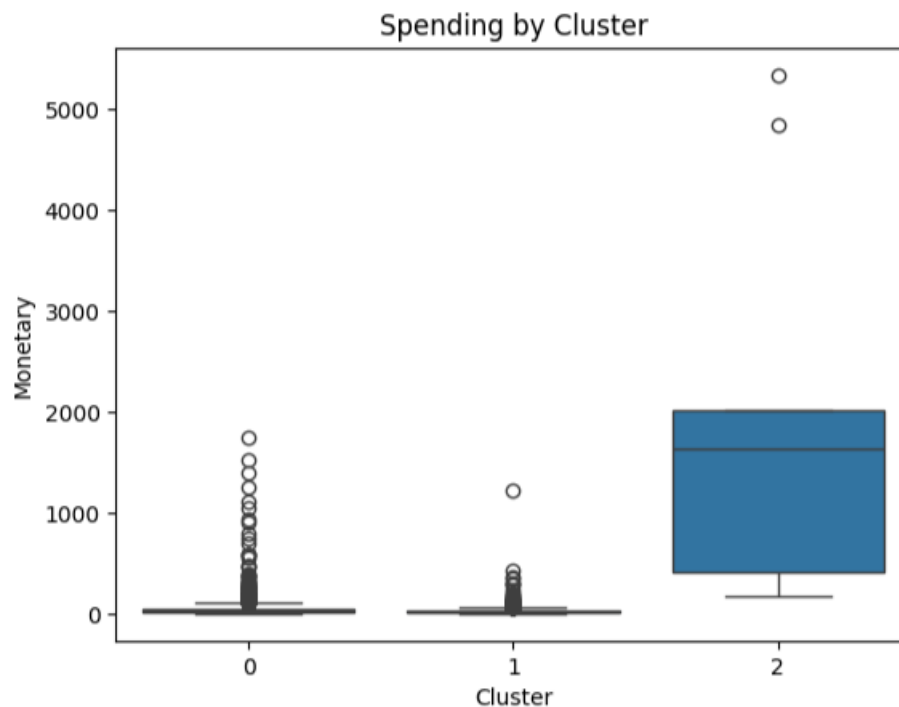
## 6. Clustering

K-Means clustering technique was applied to group customers into three segments based on their RFM values. This helps in identifying high-value, regular, and low-value customers.

- from sklearn.preprocessing import StandardScaler
- from sklearn.cluster import KMeans
- scaler = StandardScaler()
- rfm\_scaled = scaler.fit\_transform(rfm[['Recency','Frequency','Monetary']])
- kmeans = KMeans(n\_clusters=3, random\_state=123)
- rfm['Cluster'] = kmeans.fit\_predict(rfm\_scaled)

```
rfm.groupby('Cluster').mean()
```

|         | Recency    | Frequency | Monetary    |
|---------|------------|-----------|-------------|
| Cluster |            |           |             |
| 0       | 53.070370  | 2.691111  | 52.693015   |
| 1       | 253.270642 | 1.484709  | 29.180061   |
| 2       | 8.111111   | 44.000000 | 1929.121111 |



## 7. Classification

A Decision Tree classification model was used to predict customer segments based on RFM features. This demonstrates the predictive capability of the model.

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

Features and Target
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['Cluster']

Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=123)

Model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

Prediction
y_pred = model.predict(X_test)

Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", accuracy)
```

---

Model Accuracy: 0.9925558312655087

# ANALYSIS & RESULTS

The dataset was analyzed using various visualization techniques and RFM analysis to understand customer purchasing behavior. The key observations are as follows:

## 1. Customer Spending (Bar Chart)

The bar chart of top customers shows that a small number of customers contribute a large portion of total revenue. This indicates the presence of high-value customers who are important for the business.

### Interpretation:

The bar chart shows that a few customers contribute significantly to total revenue, indicating the presence of high-value customers.

## 2. Spending Distribution (Histogram)

The histogram shows that most customers have low spending, while only a few customers have very high spending. This indicates a skewed distribution where revenue is concentrated among a limited number of customers.

### Interpretation:

The histogram shows that most customers have low spending, while only a few customers have very high spending.

## 3. Sales Trend (Line Chart)

The line chart shows fluctuations in sales over time. There are periods of high and low sales, indicating changes in demand, which may be influenced by seasonal trends or customer activity.

### Interpretation:

The line chart shows fluctuations in sales over time, indicating varying demand patterns.

## 4. Customer Segmentation (Scatter Plot)

The scatter plot shows customers grouped into different clusters based on Recency and Monetary values. Each color represents a different cluster formed using K-Means clustering.

- One cluster represents high-value customers with high spending and recent activity
- Another cluster represents regular customers with moderate spending
- The third cluster represents low-value or inactive customers

**Interpretation:**

The scatter plot shows customer segmentation, where different colors represent different clusters based on purchasing behavior.

**5. Spending by Cluster (Box Plot)**

The box plot shows the distribution of customer spending across different clusters. One cluster has significantly higher monetary values, indicating high-value customers. Other clusters show lower spending levels, representing regular and low-value customers.

**Interpretation:**

The box plot shows variation in spending across clusters, with one cluster having significantly higher values.

**6. Cluster Analysis**

The clustering results clearly divide customers into three meaningful groups. This segmentation helps in identifying valuable customers and understanding different purchasing patterns.

**7. Classification Results**

A classification model was applied to predict customer segments based on RFM values. The model achieved good accuracy, indicating that customer behavior can be effectively predicted using these features. This demonstrates the practical use of machine learning in customer analysis.

Overall, the analysis reveals that customer behavior varies significantly, and a small group of customers contributes most of the revenue.

# FINDINGS

The key findings of the project are as follows:

- A small number of customers contribute a large portion of total revenue, indicating the presence of high-value customers.
- Most customers have low purchase frequency and low spending.
- Customer spending is highly skewed, with few customers showing very high monetary value.
- Sales trends show fluctuations over time, indicating varying demand patterns.
- Customers can be effectively segmented into three groups: high-value customers, regular customers, and low-value or inactive customers.
- The clustering technique successfully grouped customers based on their purchasing behavior.
- The classification model demonstrated that customer segments can be predicted using RFM values.

# CONCLUSION

This project successfully analyzed customer purchasing behavior using transaction data. By applying RFM (Recency, Frequency, Monetary) analysis, valuable insights were obtained regarding customer activity and spending patterns.

The use of data visualization techniques helped in understanding trends in customer spending and sales over time. K-Means clustering effectively segmented customers into different groups such as high-value, regular, and low-value customers. Additionally, the classification model demonstrated that customer segments can be predicted using RFM values.

The results show that a small number of customers contribute significantly to total revenue, while the majority of customers have lower engagement. This highlights the importance of identifying and targeting high-value customers.

Overall, this project demonstrates how data science techniques can be used to analyze customer behavior and support business decision-making. These insights can help businesses improve customer retention, design targeted marketing strategies, and increase profitability.